

# Blender & 3D Modelling

## 4-Session Plan

A hands-on curriculum for the Game Design Chapter | All levels welcome | Blender is free

S1

### Welcome to Blender — navigating & thinking in 3D

Week 1 | 90 min

BEGINNER

#### WHAT YOU'LL COVER

- Blender interface – viewport, outliner, properties
- Navigation: orbit, pan, zoom, numpad shortcuts
- Object mode vs edit mode
- Basic transforms: Grab / Rotate / Scale (G/R/S)
- Adding & deleting meshes (Shift+A)

#### HANDS-ON ACTIVITY

- Build a game prop (crate or barrel) using primitives & boolean cuts
- Organise the outliner, export as .fbx
- Group discussion: 3D objects in your favourite games

#### RESOURCES

blender.org – Blender (free & open source 3D creation software) | Blender Guru Beginner Series | Blender Fast Track | docs.blender.org – Official manual

#### TAKE-HOME

Recreate one prop from a game of your choice – bring it to session 2

S2

### Modelling & topology — building game-ready assets

Week 2 | 90 min

FUNDAMENTALS

#### WHAT YOU'LL COVER

- Edge loops, vertex merging, face fills
- Loop cuts, extrude, inset, bevel
- Good topology – why it matters for games
- Subdivision surface modifier
- Low poly vs high poly – engine constraints

#### HANDS-ON ACTIVITY

- Model a low-poly character head or stylised helmet
- Peer critique – share screens, discuss topology
- Mirror modifier intro – model only half

#### RESOURCES

YouTube: Grant Abbitt – low poly character topology for Beginners | sketchfab.com/explore – Reference models – Free assets

#### TAKE-HOME

Complete the character head and add simple colour materials

S3

## UV unwrapping &amp; texturing — making it look like a real game

Week 3 | 90 min

INTERMEDIATE

## WHAT YOU'LL COVER

- What is UV unwrapping and why it exists
- Seams, smart UV project, UV editor
- Applying image textures in the shader editor
- Base colour, roughness, metallic – PBR basics
- Baking normals from high poly to low poly

## HANDS-ON ACTIVITY

- UV unwrap your prop or character head from session 2
- Paint a texture in Krita or use a free PBR set from Poly Haven
- Render a beauty shot using Eevee

## RESOURCES

YouTube: UV Unwrapping – Blender.org – Free texture painting: cg.com – Free PBR textures: Stylized Station

## TAKE-HOME

Fully textured asset ready to import into Godot or Unity for session 4

S4

## Export, import &amp; showcase — getting into the game engine

Week 4 | 90 min

CAPSTONE

## WHAT YOU'LL COVER

- Export formats: .fbx, .glTF, .obj – when to use each
- Importing into Godot 4 or Unity
- Setting up materials in the engine vs Blender
- Basic rigging – armatures and bones
- Where next: animation, sculpting, character rigging

## HANDS-ON ACTIVITY

- Export your asset from sessions 1-3 into a shared scene
- Mini showcase – screenshot your asset in the engine
- Group vote on next topic

## RESOURCES

godotengine.org – Godot (free) | unity.com – Unity (free personal) | YouTube: GDQuest – Godot tutorials | artstation.com/learning – ArtStation Learning

## TAKE-HOME

One complete, textured, engine-ready 3D asset per member – post to Discord

## Always-on resources for the chapter

- Blender Artists Community forum
- CG Cookie – structured Blender courses
- GDC Vault – free conference talks
- Brackeys (YouTube) – Unity/Godot pipeline
- r/blender & r/gamedev (Reddit)
- 80.lv – industry art breakdowns
- OpenGameArt.org – free game assets
- ArtStation Learning – pro tutorials